

Analyzing Student Success

Ctrl Alt Defeat

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How does success across learnosity
question types influence students'
overall success?

Hypothesis and Thoughts

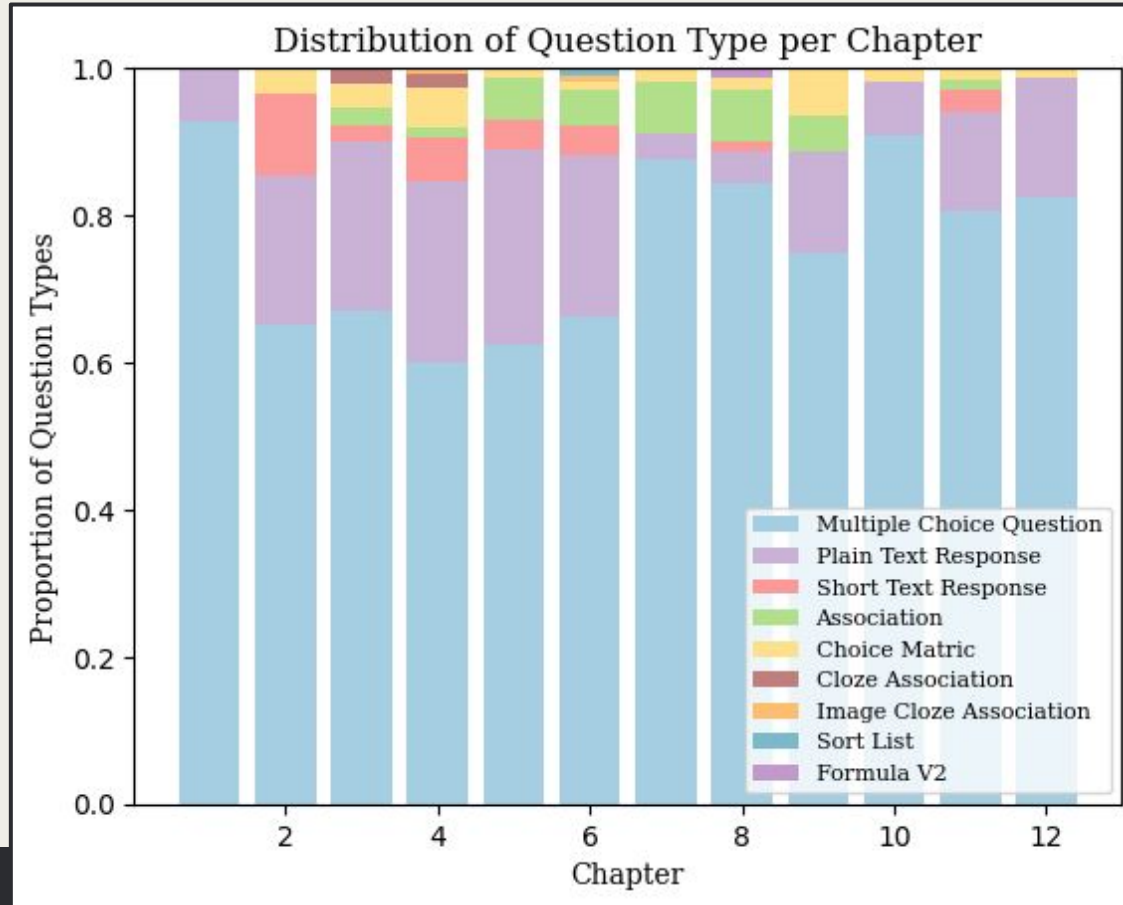
Hypothesis: There are learnosity question types that certain groups of students succeeded at more than others.

In order to make our data analysis simple, we only used information from the *College / Statistics and Data Science (ABC)* book.

1. To test our hypothesis, we first merged each csv by student_id, so we had all information for that student in one location.
2. We then grouped them into 4 groups corresponding to their percentile based on EOC response correctness across chapters. (25th, 50th, 75th, 100th).
3. We then analyzed the EOC averages of each learnosity question type in hopes of finding patterns based on students' percentile.

We'll use this information to make recommendations to CourseKata on how best to work towards their goal of improving statistical learning.

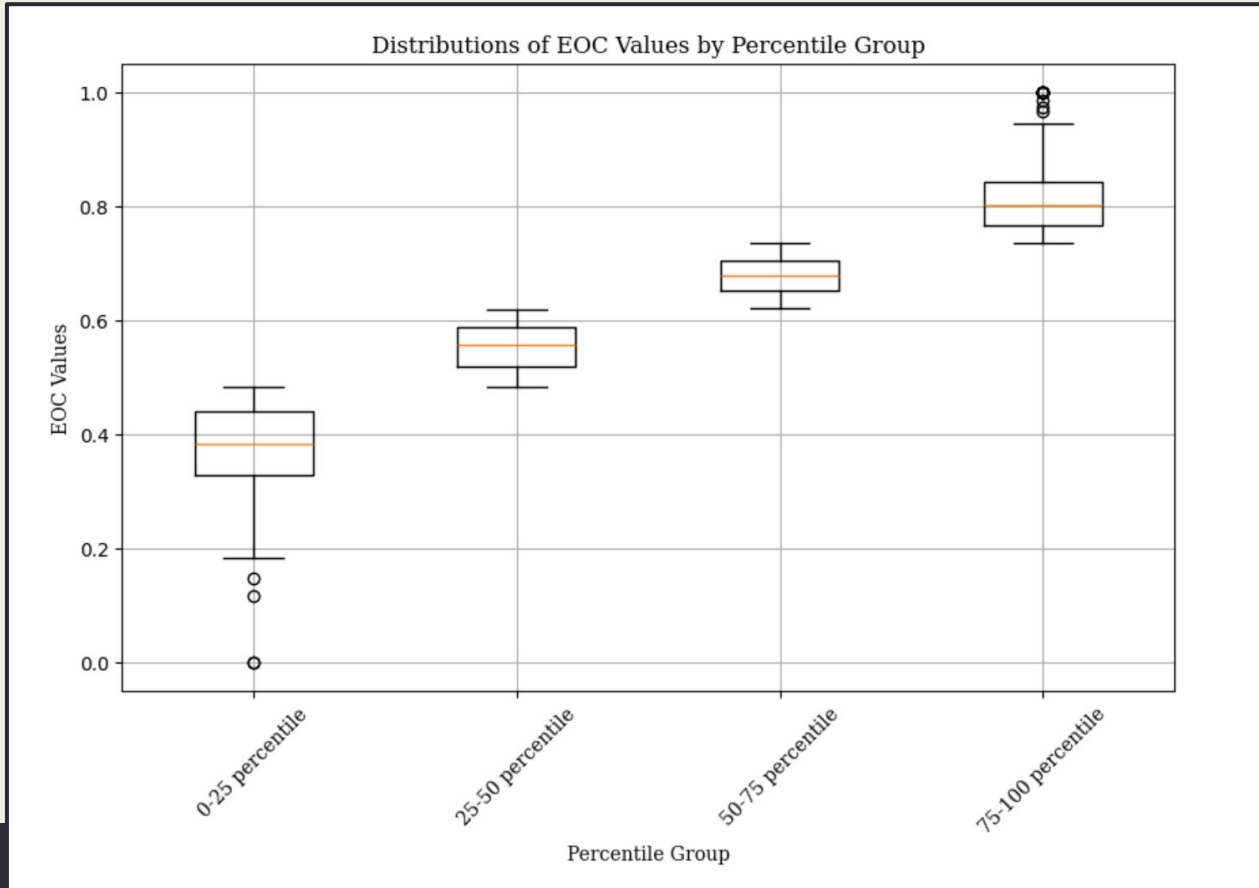
Graph 1: Distribution of Question Type per Chapter



Observations:

- Multiple choice questions make up the majority of questions in each chapter
- Long text response have the second highest probability
- Ordering problems are only present in one chapter

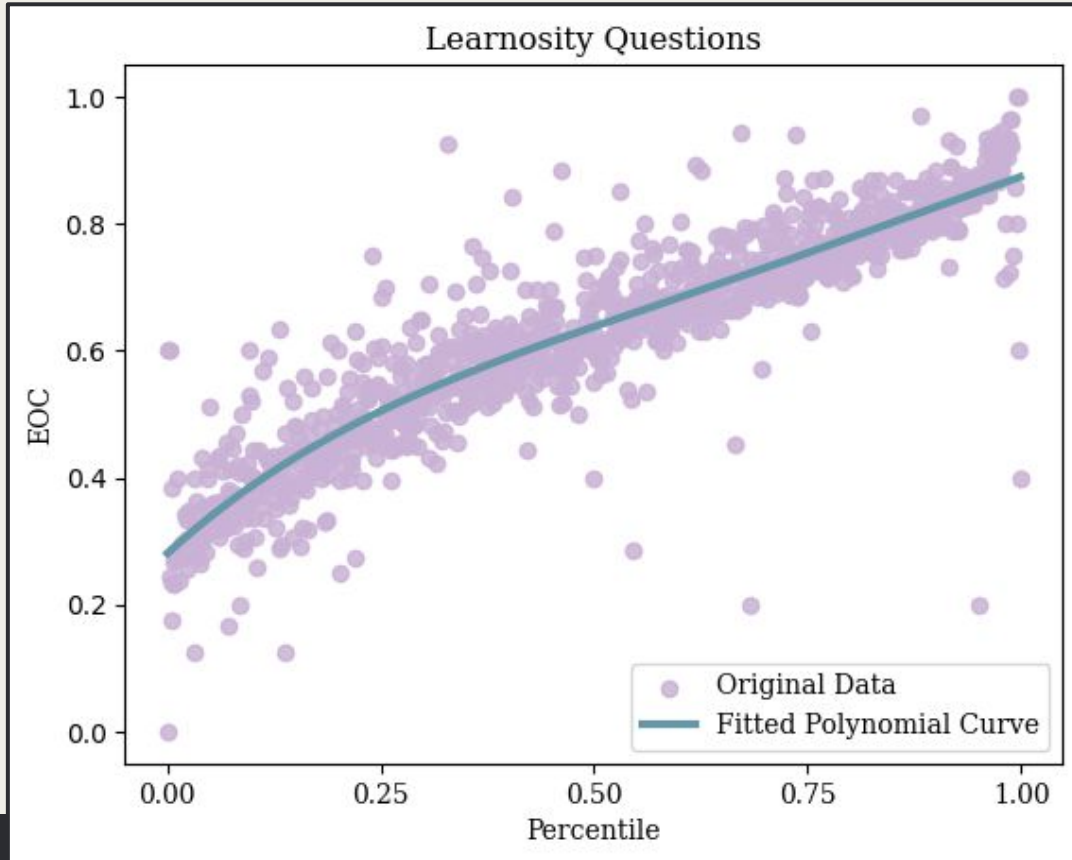
Graph 2: Distribution of EOC Values by Percentile Group



Observations:

- 25%-50% and 50%-75% have the least variations in scores
- 0%-25% has the most amount of variation
- 75%-100% has the most amount of outliers

Graph 3: Percentile vs EOC for total Learnosity Questions

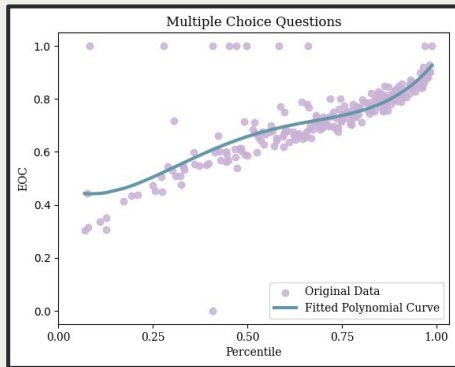


Observations:

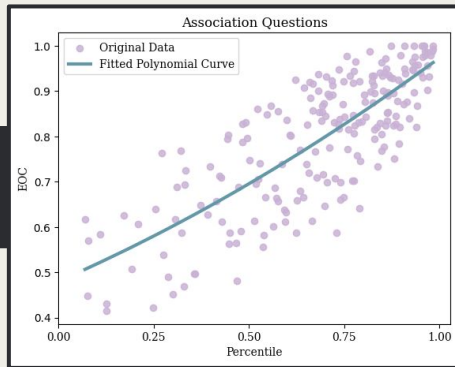
- Fairly positive linear correlation
- We can see a slight incline in performance within the 25th percentile

Graph Group: EOC for Type of Learnosity Question by Percentile

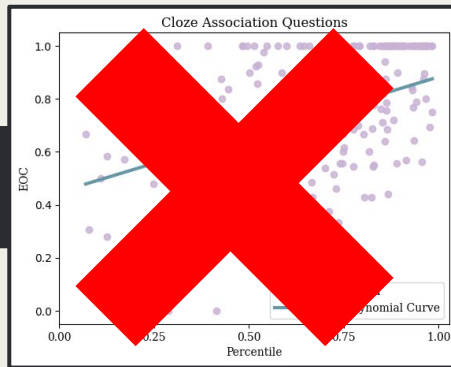
785 MC Questions



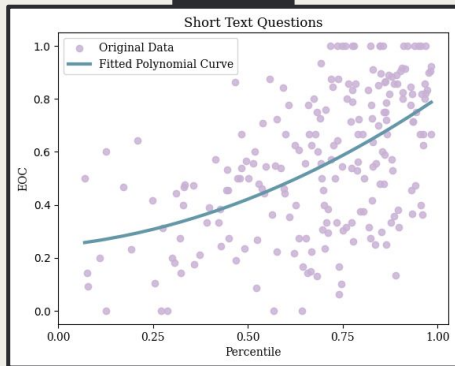
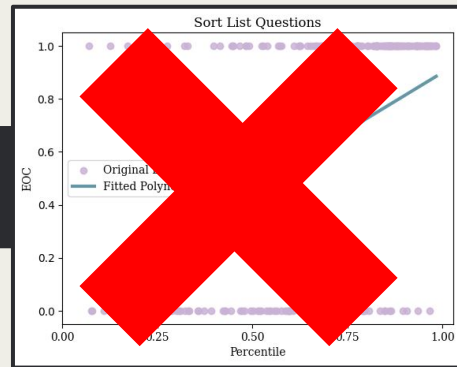
28 Association Questions



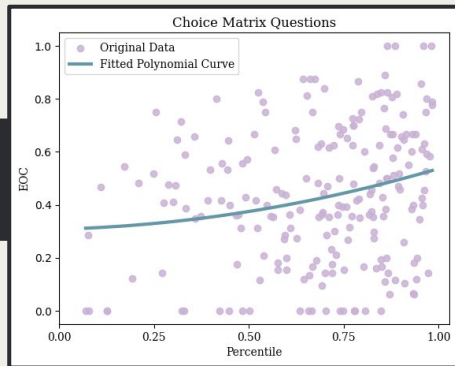
5 CA Questions



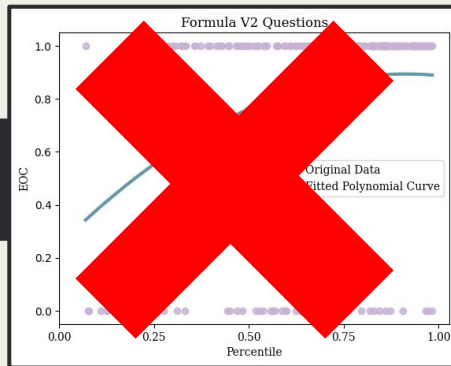
1 SL Question



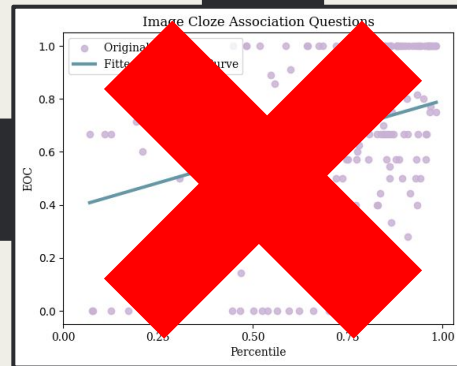
35 ST Questions



45 CM Questions

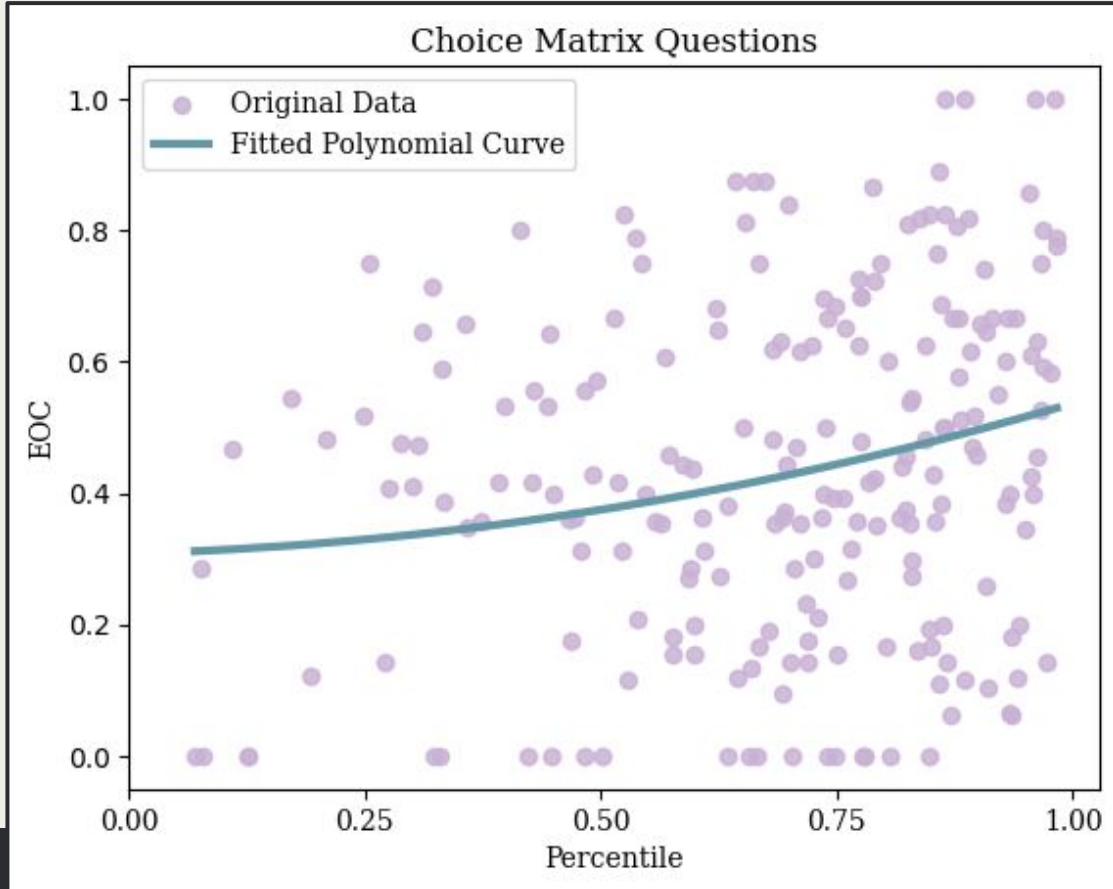


1 FV2 Question



2 ICA Questions

Graph 4: Choice Matrix Questions



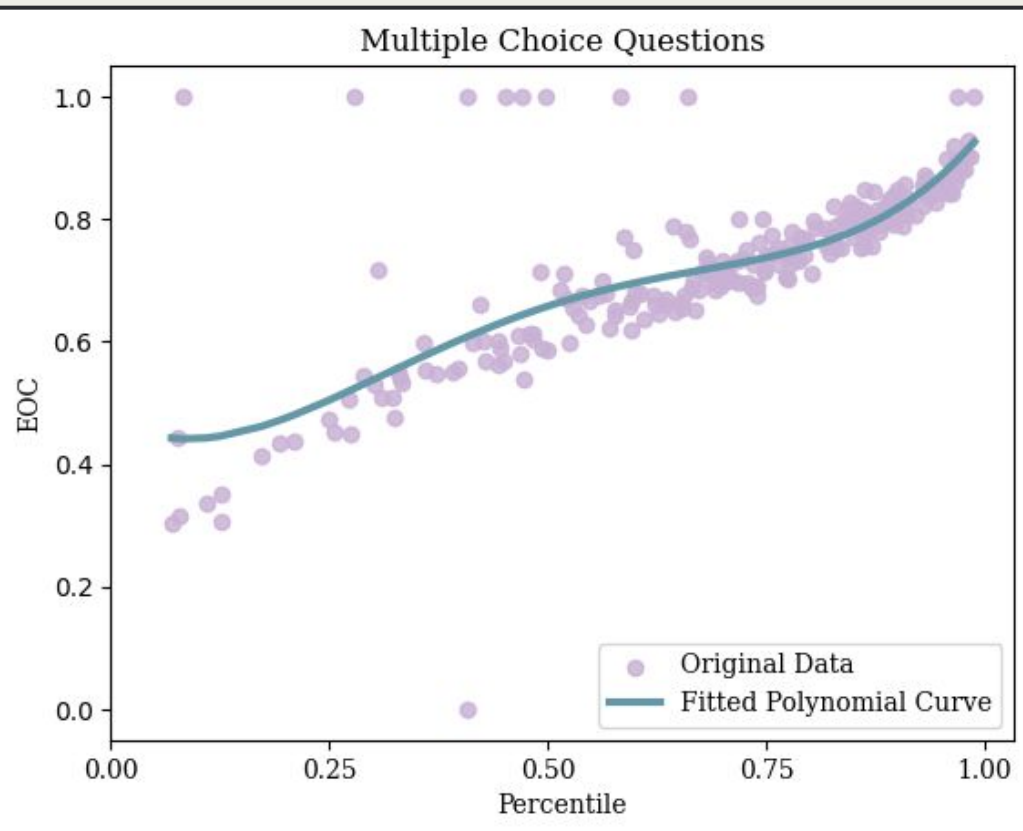
Observations:

- Little difference between the average Choice Matrix EOCs of students who performed better overall and those who did not

Which of these are values and which are variables?

		Value	Variables	Neither
A	Condition	☐	☐	☐
B	Wt2	☐	☐	☐
C	45	☐	☐	☐
D	Housekeeper	☐	☐	☐
E	Uninformed	☐	☐	☐
F	135.8	☐	☐	☐

Graph 5: Multiple Choice Questions



Observations:

- Fairly positive linear correlation
- Steeper incline within 0%-25% and 75%-100%
- Steadies off(flatter) in the middle 50%

In this snippet of data, there are 6 ages listed: 35, 45, 52, 29, 38, and 39. Does this mean there are six variables for Age?

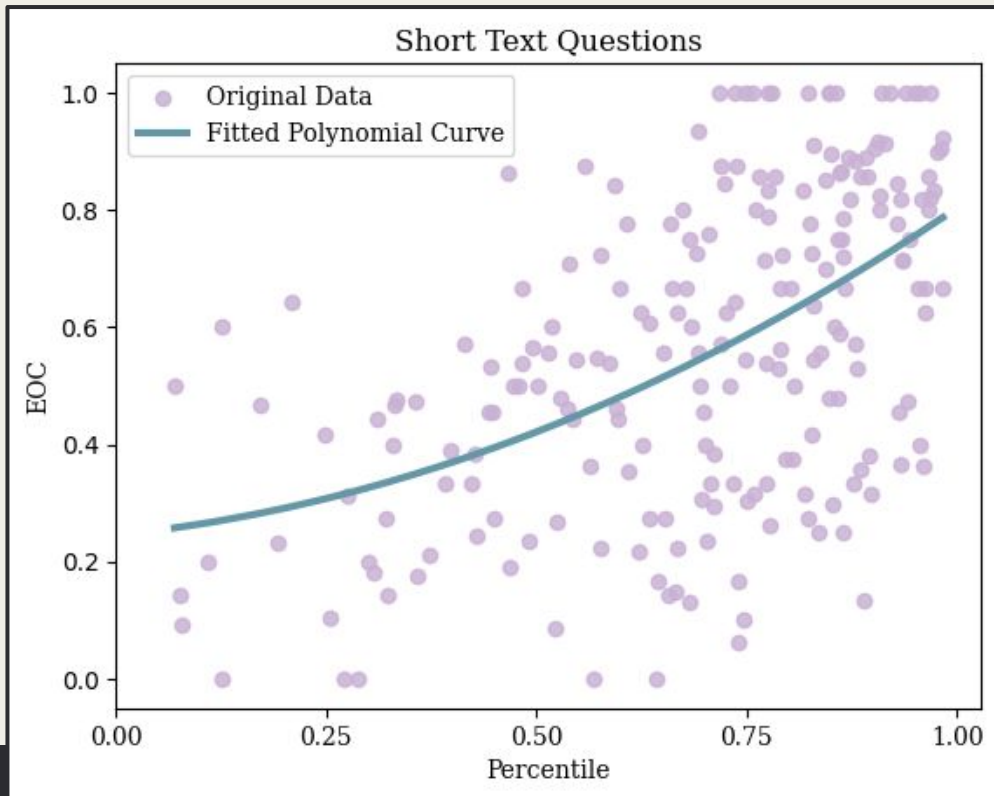
	Condition	Age	Wt	Wt2
1	Uninformed	35	136	135.8
2	Uninformed	45	162	161.8
3	Informed	52	117	116.8
4	Informed	29	184	182.8
5	Uninformed	38	134	136.6
6	Informed	39	189	183.2

A Yes

B No

C I don't know.

Graph 6: Short Text Questions



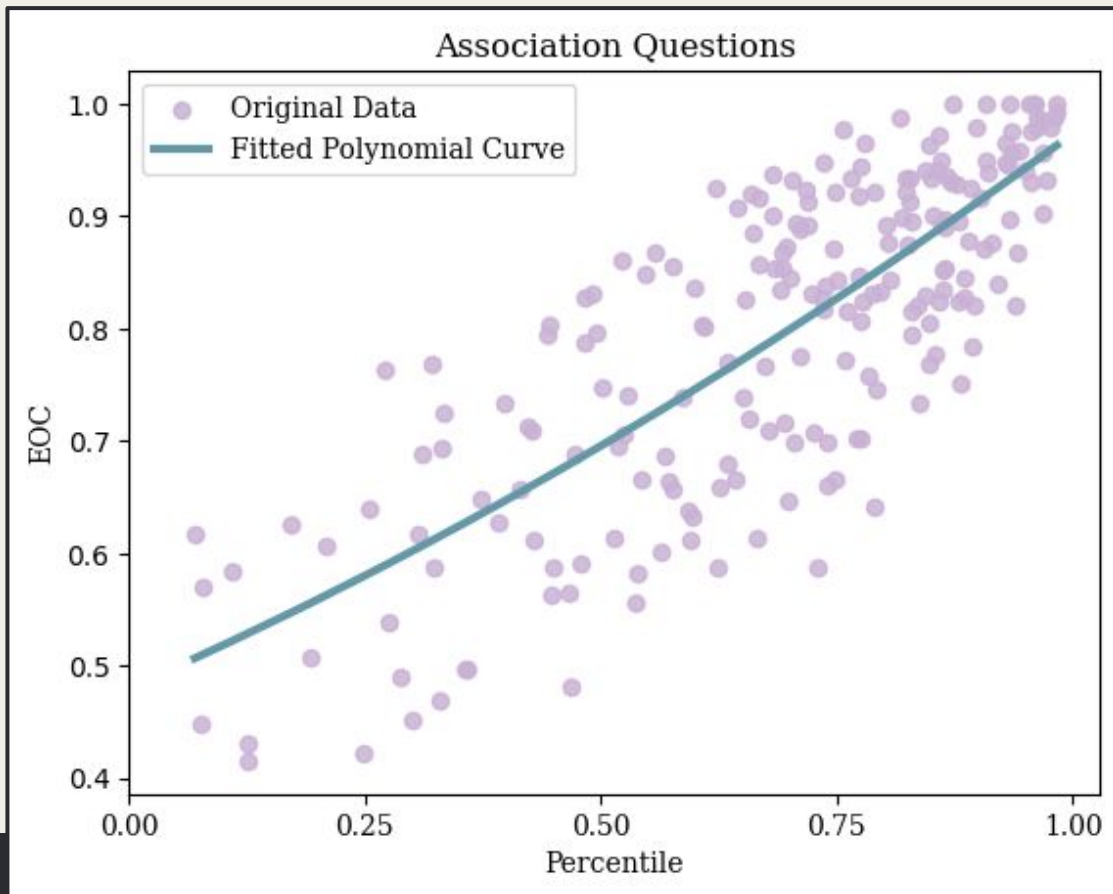
Observations:

- Somewhat positive linear relationship between percentile and EOC
- Data is very scattered throughout the graph

	Condition	Age	Wt	Wt2
1	Uninformed	35	136	135.8
2	Uninformed	45	162	161.8
3	Informed	52	117	116.8
4	Informed	29	184	182.8
5	Uninformed	38	134	136.6
6	Informed	39	189	183.2

Notice the numbers 1 through 6 down the left column. What do these numbers represent?

Graph 7: Association Questions



Observations:

- Students EOC increases as rank percentile increases
- Higher rank percentile shows increased ability to answer association questions

Which features of the boxplot go with which statistical summary?

The lowest point of the bottom whisker	
The top line of the box (this is called the top hinge)	
The bottom line of the box (this is called the bottom hinge)	
The height of the box	
The highest point of the top whisker	
The middle line of the box	

Statistical summaries:

- Max (Q4) = 196
- Q3 = 161.5
- IQR = Q3 - Q1 = 31.5
- Median (Q2) = 145
- Min (Q0) = 90
- Q1 = 130

Correlations Between Question Type and User Satisfaction

We compared the proportion of questions present within a chapter to the user response in the pulse check:

Correlation to Engagement/Idle ratio

Cost	-0.643859
Expectancy	0.286992
Intrinsic Value	0.607907
Utility Value	0.479783
mcq	-0.305935
plaintext	0.366346
shorttext	0.431620
association	-0.753733
choicematrix	0.303040
clozeassociation	0.551106

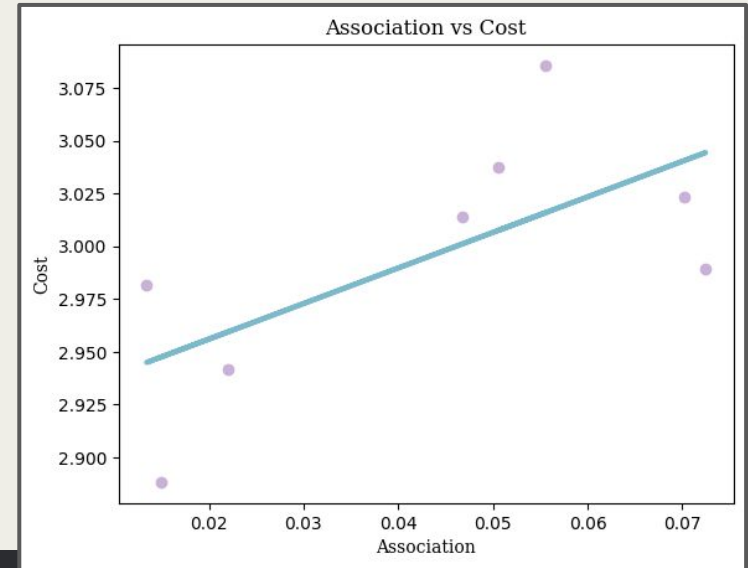
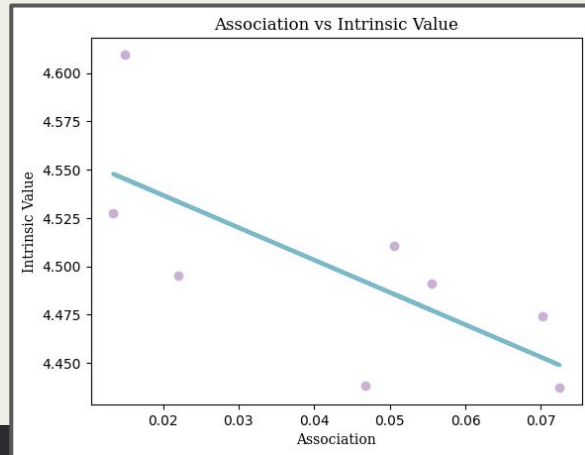
Values to question type

	Cost	Expectancy	Intrinsic Value	Utility Value
mcq	0.096503	0.011247	-0.182508	0.090000
plaintext	-0.031068	-0.006599	0.259315	0.002273
shorttext	-0.649136	0.593800	0.668936	0.486908
association	0.597050	-0.529962	-0.749443	-0.698855
choicematrix	-0.170871	-0.097443	-0.093323	-0.225157
clozeassociation	-0.002401	-0.441795	-0.089914	-0.280534
engagement_to_distracted	-0.643859	0.286992	0.607907	0.479783

Association Questions

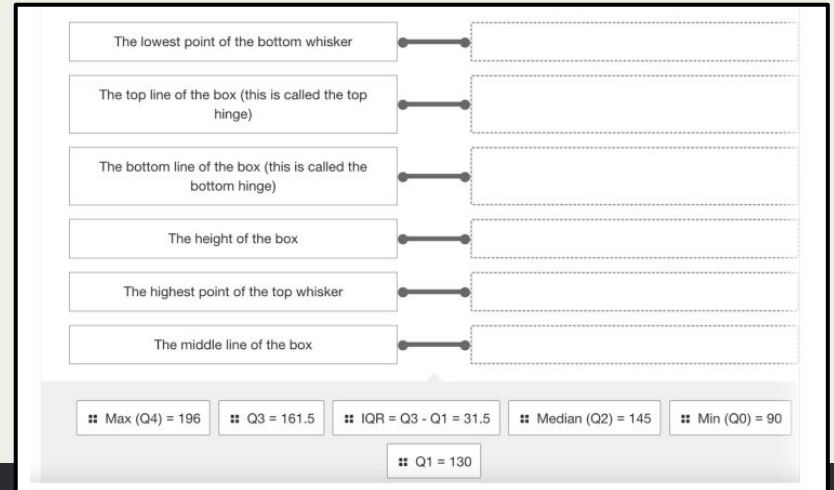
- Students likely perceive the Association questions as time intensive, unengaging, and with little value.
- Ability to answer association questions is a strong indicator of a student's knowledge

	Cost	Expectancy	Intrinsic Value	Utility Value
association	0.597050	-0.529962	-0.749443	-0.698855



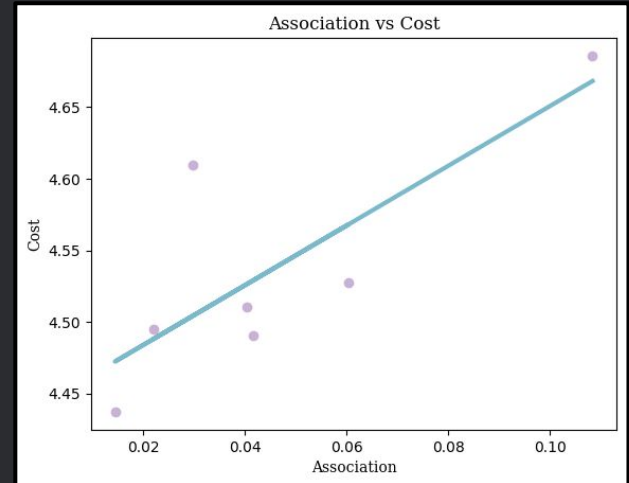
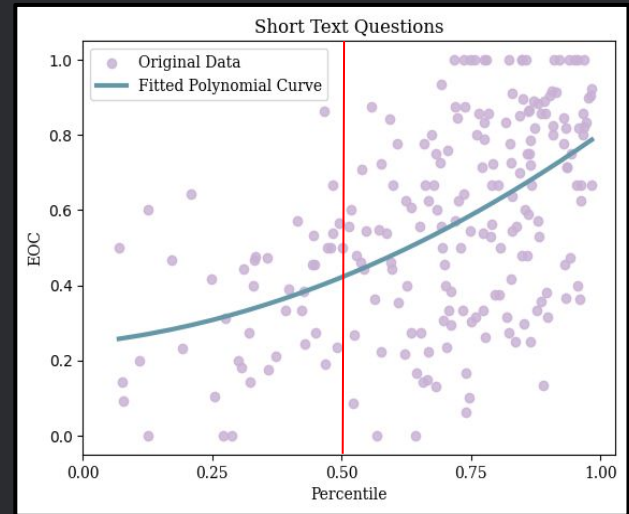
Solution to Association Questions

- Use to evaluate a user's capacity
- Reduce quantity in practice and homework



Short Text Questions

- Weaker students answer a fewer number of these questions
- **Solution:** Increase engagement by naturally guiding students into the question
- Chapters with a larger number of these questions are perceived more positive across the board
- **Solution:** Increase the number of short test questions. Allow students to explain their confusion for partial credit



Our Recommendations

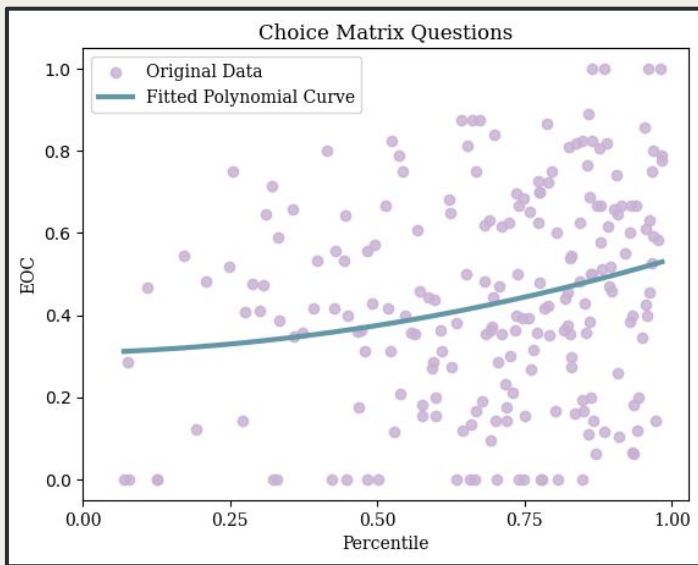
Volunteer Response Bias

Fully integrate an **“I don’t know”** option.

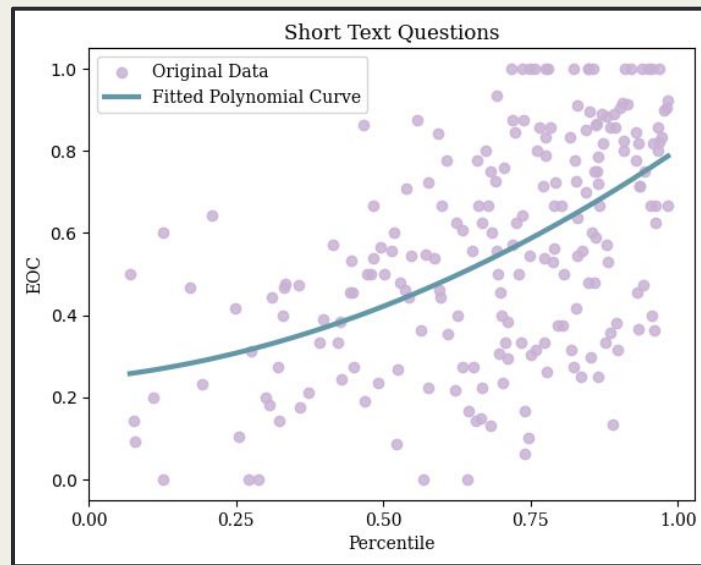
Keep students from getting ahead of themselves.

Not all questions are created equal.

Some questions-types are **poor indicators of student success.**



These questions should be **altered** or **replaced** with questions that are more helpful.



The End

Thank you for listening!